TSNM (Touch Sensing Note Memory) MKII 4U v1.0

BILL OF MATERIALS / THROUGH-HOLE (THT) VERSION / www.doboz.audio

Qty	Parts	Value	Description	Mouser	TME	Alternatives
11	**R8-R17, R35**	100R	resistor, <=5%, 600mW	594-MRS25000C1000FCT	MF0207FTE52-100R YAGEO	MBB02070C1000FCT00, MRS25000C1000FCT00
7	**R5, R6, R7, R18-R21**	470R	resistor, <=5%, 600mW	594-MRS25000C4700FCT	MF0207FTE52-470R YAGEO	MBB02070C4700FCT00, MRS25000C4700FC100
3	**R26, R30, R38**	1k	resistor, <=5%, 600mW	594-MRS251K1%TR	MF0207FTE52-1K YAGEO	MBB02070C1001FCT00, MRS25000C1001FCT00
3	**R25, R29, R37**	10k	resistor, <=5%, 600mW	594-MRS25000C1002FCT	MF0207FTE52-10K YAGEO	MBB02070C1002FCT00, MRS25000C1002FCT00
3	**R27, R28, R36**	20k	resistor, <=5%, 600mW	594-MRS25000C2002FCT	MF0207FTE52-20K YAGEO	MBB02070C2002FCT00, MF006FF2002A50
2	**R33, R34**	47k	resistor, <=5%, 600mW	594-MRS25000C4702FCT	MF0207FTE52-47K YAGEO	MBB02070C4702FCT00, MRS25000C4702FCT00
4	**R1, R2, R3, R4**	100k	resistor, <=5%, 600mW	594-MRS25000C1003FCT	MF0207FTE52-100K YAGEO	MBB02070C1003FCT00, MRS25000C1003FCT00
1	**R32**	1k	resistor, <=0.1%, 1/4W, 15PPM/C	279-YR1B1K0CC	MBB0207VD1001BC100 VISHAY	
1	**R22**	30k1	resistor, <=0.1%, 1/4W, 15PPM/C	279-YR1B30K1CC	MRA020730K1BTA15 ARCOL	
1	**R31**	33k	resistor, <=0.1%, 1/4W, 15PPM/C	279-YR1B33K2CC	MF006BB3302A10 ROYAL OHM	
2	**R23, R24**	100k	resistor, <=0.1%, 1/4W, 15PPM/C	279-YR1B100KCC	MBB0207VD1003BC100 VISHAY	
1	**C10**	22p	capacitor, >=25V, C0G, pitch: 5mm	594-K220J15C0GF5TL2	CM-22 SR PASSIVE	
2	**C15, C16**	220p	capacitor, >=25V, C0G, pitch: 5mm	594-K221J15C0GF5UH5	CM-220 SR PASSIVES	
12	**C1-C9, C11, C12, C14**	100n	capacitor, >=25V, pitch: 5mm	505-MKS2D031001A00MC	R82DC3100DQ50J KEMET	
1	**C13**	1u	capacitor, >=25V, pitch: 5mm, height <1cm	505-MKS2B041001C00MF	B32529C0105K289 EPCOS	
5	**C19-C23**	22u	capacitor, electrolytic, >=35V, max. Ø6.3mm	667-EEA-GA1V220B	UMT1V220MDD1TP NICHICON	EEAGA1V220 PANASONIC
2	**C17, C18**	47u	capacitor, electrolytic, >=35V, max. Ø6.3mm	667-EEU-FC1V470	ULD1V470MED NICHICON	RD1V476M05011BB SAMWHA
2	**IC1, IC2**	74HC595	shift register, DIP16	595-SN74HC595N	SN74HC595N TEXAS INSTRUMENTS	
1	**IC3**	DAC8560	16-bit DAC, VSSOP8	595-DAC8560ICDGKT	-	
1	**IC4**	OPA277	operational amplifier, DIP8	595-OPA277PA	OPA277PA TEXAS INSTRUMENTS	TL071
1	**IC5**	TL074	operational amplifier, DIP14	595-TL074CN	TL074ACN TEXAS INSTRUMENTS	
1	**IC6**	78L33	voltage regulator, 3.3V, 0.1A, TO92	511-L78L33ACZ	L78L33ABZ-AP STMICROELECTRONICS	
1	**IC7**	R-78E5.0-0.5	DC/DC converter, 5V, 500mA, SIP3	919-R-78E5.0-0.5	AMSR-7805-NZ AIMTEC	
2	**Q1, Q2**	2N3904	transistor, NPN, bipolar, 40V, 0.2A, TO92	610-2N3904	2N3904BU ON SEMICONDUCTOR	
2	**D1, D2**	1N4148	small signal switching diode, DO35	512-1N4148	1N4148 DIOTEC SEMICONDUCTOR	
2	**D3, D4**	1N5819	diode, Schottky rectifying, DO41	511-1N5819	1N5819 DC COMPONENTS	1N4001
1	**L1**	33uH	inductor, 33uH, 200mA	542-78F330-RC	MICC-330K-02 FASTRON	
3	**L2, L3, L4**	BL02RN1R2M2B	ferrite bead	81-BL02RN1R2M2	BL02RN1R2M2B MURATA	434-BEAD/4-900R-02, 807-4211R-28TR, 10 Ohm resistor
1	**ENC1**	Encoder	incremental encoder, 20imp/revol	858-EN11-HSM1BF20	EC11E20-20P20C-SW SR PASSIVES	
1	**ENC1 (knob)**	knob	encoder knob for Ø6mm shaft	450-BA600	CL17094 CLIFF + K85-BLK	thonk.co.uk
2	**SW1, SW2**	KS01-BL	keypad switch with LED	611-D6RLRDF1LFS	KS01-BLV-1 HIGHLY	comes with the kit
2	**MATRIX1, MATRIX2**	5x7 LED matrix	display, LED matrix, 5x7, com. cathode	859-LTP-747KR	FYM-7571AUHR-11 FORYARD	
1	**TEENSY 3.2**	-	microcontroller breakout board	474-DEV-13736	-	pjrc.com, TSNM will not work with Teensy 4.0!
1	**MPR121 BOARD**	-	capacitive touch sensor breakout		eBay	485-1982
2	**IC1, IC2 (socket)**	IC socket	DIP socket, 16 position	575-110433161	DS1009-16AT1NX-0A2 CONNFLY	
1	**IC4 (socket)**	IC socket	DIP socket, 8 position	575-144308	DS1009-08AT1NX CONNFLY	
1	**IC5 (socket)**	IC socket	DIP socket, 14 position	575-11044314	DS1009-14AT1NX-0A2 CONNFLY	
1	**TEENSY 3.2 (socket)**	socket	female socket, cut to length!	517-929974-01-30-RK	-	
1	**MPR121 BOARD (socket)**	socket	female socket, cut to length!	517-929974-01-20-RK	-	
1	**J7 (PCB)**	1x15 pin	female socket, 2.54mm, height: 7mm	437-8018702010001101	BL5.36Z FISCHER ELEKTRONIK	
1	**J7 (panel)**	1x15 pin	SMD male header, 2.54mm	538-87898-2026	PH1-40-UA-SMT-B ADAM TECH	
1	**POWER CONNECNTOR**	EDAC connector	EDAC card edge connector, 10 positions	587-306-50-010	-	
2	**CLOCK, RESET**	banana jack	banana jack, orange	530-108-0906-1	-	samodular.com, thonk.co.uk
1	**PITCH**	banana jack	banana jack, blue	530-108-0910-1	-	samodular.com, thonk.co.uk
1	**PRESSURE**	banana jack	banana jack, violet	530-108-0912-1	-	samodular.com, thonk.co.uk
2	**PULSE 1, PULSE 2**	banana jack	banana jack, red	530-108-0902-1	-	samodular.com, thonk.co.uk
2	**spacer**	-	10mm spacer, male/female	710-971100365	TFM-M3X10/DR234	710-971100354
2	**screw**	-	M3 screw	534-9191-3	1020498 BOSSARD	
2	**nut**	-	M3 nut	534-4688	BOSSARD 1089315	144-04M030050HNDIN34